

Technical Document #898: Retrieval Augmented Generation - Comprehensive Overview

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Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.
- Re-ranking models score retrieved documents for relevance to the query.
- Chunking splits documents into smaller segments for embedding.